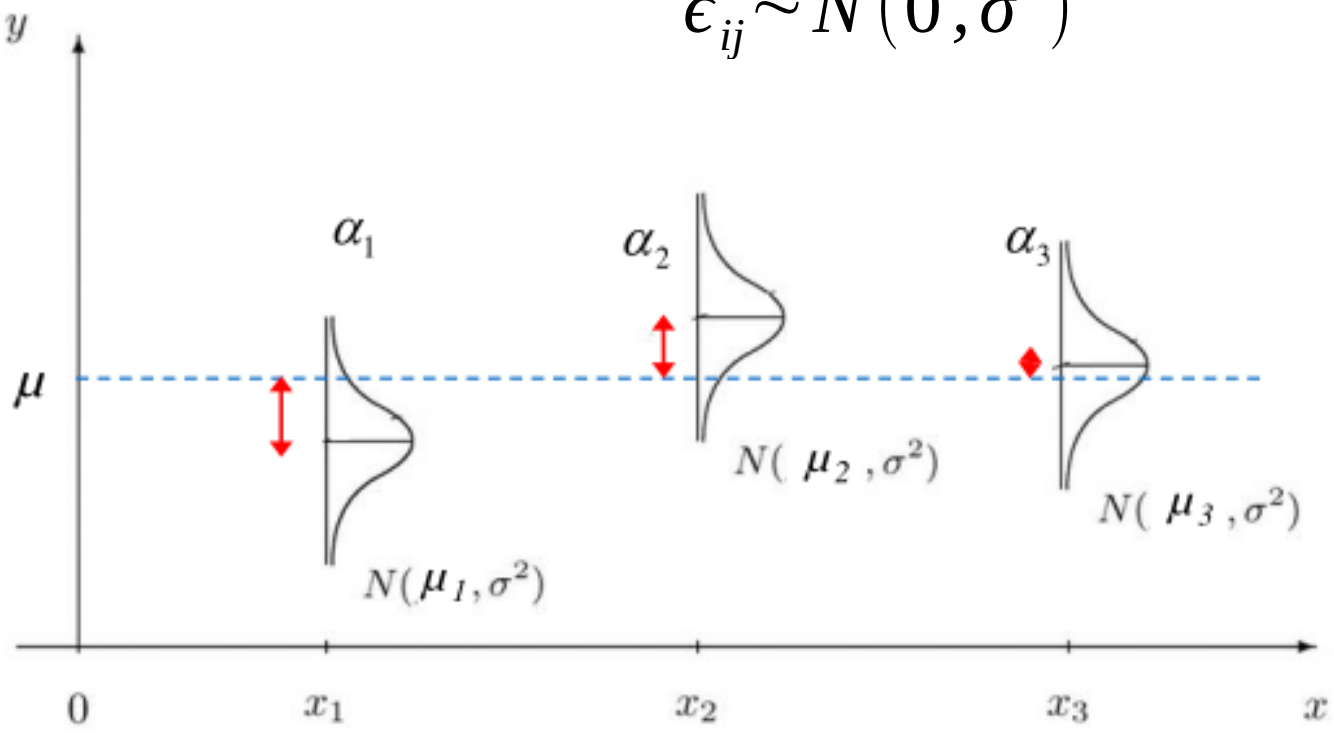
ANOVA: comparación de tres o más medias

	(X_1)	(X_2)	(X_3)
	Toxin 1	Toxin 2	Toxin 3
(Y)	28	33	18
	23	36	21
	14	34	20
	27	29	22
		31	24
		34	

Daño al hígado por causa de tres toxinas diferentes

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

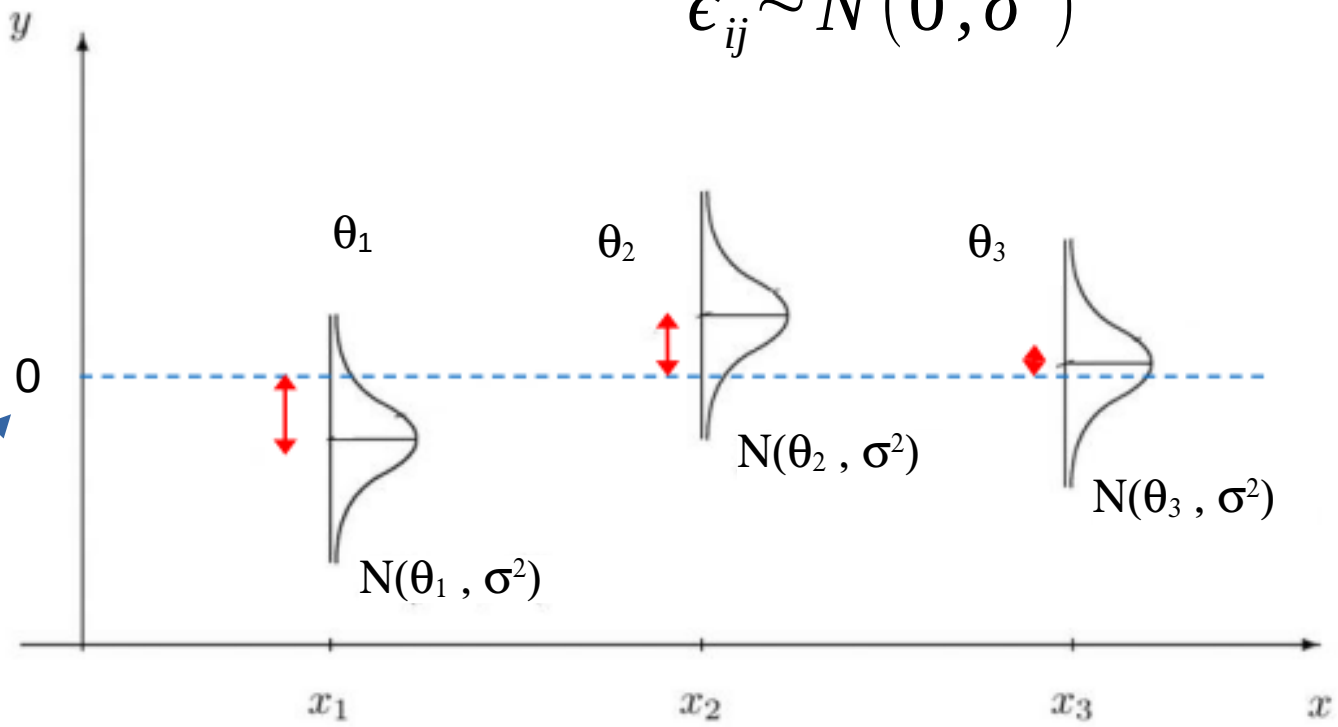


ANOVA: comparación de tres o más medias

Treatments			
1	2	3	...
y_{11}	y_{21}	y_{31}	...
y_{12}	y_{22}	y_{32}	...
⋮	⋮	⋮	...
		y_{3n_3}	
y_{1n_1}			
	y_{2n_2}		

$$Y_{ij} = \theta_i + \epsilon_{ij}$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$



Restamos valor medio global

Hay diferencias entre los grupos?

La estimación puntual es como antes: máxima verosimilitud

- estimación de θ_i a través de valor medio muestral de cada grupo
- estimación de σ^2 a través de varianza muestral

La pregunta: cómo hacemos para decidir si hay diferencias significativas entre las estimaciones de los valores medios? (contraste de hipótesis, intervalos de confianza)

Necesitamos algunas hipótesis sobre la distribución de los epsilon

Oneway ANOVA assumptions

Random variables Y_{ij} are observed according to the model

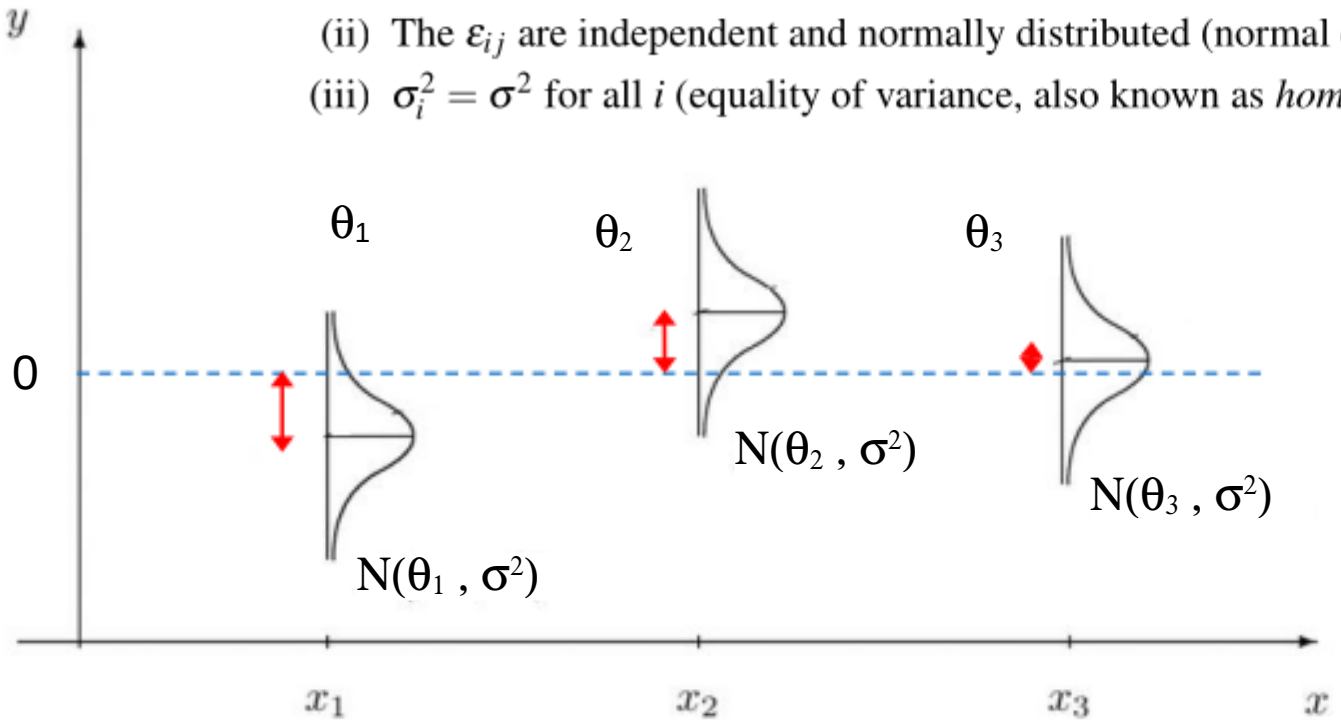
$$Y_{ij} = \theta_i + \varepsilon_{ij}, \quad i = 1, \ldots, k, \quad j = 1, \ldots, n_i,$$

where

(i) $\mathrm{E}\varepsilon_{ij} = 0, \mathrm{Var}\,\varepsilon_{ij} = \sigma_i^2 < \infty$, for all i, j. $\mathrm{Cov}(\varepsilon_{ij}, \varepsilon_{i'j'}) = 0$ for all i, i', j, and j' unless $i = i'$ and $j = j'$.

(ii) The ε_{ij} are independent and normally distributed (normal errors).

(iii) $\sigma_i^2 = \sigma^2$ for all i (equality of variance, also known as *homoscedasticity*).



Contraste de hipótesis ANOVA de nivel α

Comparar la varianza intergrupo con la varianza intragrupo: si la primera es mayor, entonces los grupos son distinguibles

$$H_0: \quad \theta_1 = \theta_2 = \cdots = \theta_k \qquad \text{versus} \qquad H_1: \quad \theta_i \neq \theta_j \text{ for some } i, j,$$

we reject H_0 if

(varianza intergrupo)

(varianza intragrupo)

$$\frac{\sum_{i=1}^{k} n_i \left((\bar{Y}_{i\cdot} - \bar{\bar{Y}}) \right)^2}{S_p^2} > (k-1) F_{k-1,N-k,\alpha}. \tag{11.2.14}$$

This rejection region is usually written as

$$\text{reject } H_0 \text{ if } F = \frac{\sum_{i=1}^{k} n_i \left((\bar{Y}_{i\cdot} - \bar{\bar{Y}}) \right)^2 / (k-1)}{S_p^2} > F_{k-1,N-k,\alpha}$$

and the test statistic F is called the *ANOVA F statistic*.

Varianza intragrupo: “varianza pooleada”
(pooled variance)

$$S_p^2 = \frac{1}{N-k}\sum_{i=1}^{k}(n_i-1)S_i^2 = \frac{1}{N-k}\sum_{i=1}^{k}\sum_{j=1}^{n_i}(Y_{ij}-\bar{Y}_{i\cdot})^2. \qquad (11.2.5)$$

Estadístico F: cociente de dos variables independientes, cada una con distribución chi2

$$\frac{\sum_{i=1}^{k} n_i\left((\bar{Y}_{i\cdot}-\bar{\bar{Y}})\right)^2}{S_p^2} > (k-1)F_{k-1,N-k,\alpha}$$

$$Y_{ij} = \theta_i + \epsilon_{ij}$$
$$\epsilon_{ij} \sim N(0, \sigma^2)$$

ANOVA table for oneway classification

Source of variation	Degrees of freedom	Sum of squares	Mean square	F statistic
Between treatment groups	$k-1$	SSB = $\sum n_i(\bar{y}_i - \bar{\bar{y}})^2$	MSB = SSB$/(k-1)$	$F = \frac{\text{MSB}}{\text{MSW}}$
Within treatment groups	$N-k$	SSW = $\sum\sum(y_{ij} - \bar{y}_{i\cdot})^2$	MSW = SSW$/(N-k)$	
Total	$N-1$	SST = $\sum\sum(y_{ij} - \bar{\bar{y}})^2$		

Daño al hígado:

Toxin 1	Toxin 2	Toxin 3
28	33	18
23	36	21
14	34	20
27	29	22
	31	24
	34	

Source of variation	Degrees of freedom	Sum of squares	Mean square	F statistic
Treatments	3	995.90	331.97	26.09
Within	15	190.83	12.72	
Total	18	1,186.73		



p=0.000003